

Remote Sensing for Heatwave Observations

Assessing the summer 2022 heatwave in Capo Comino using the data collected from Sentinel-1 and Sentinel-2 using Google Earth Engine.

Mentor: Maria Ruchringer

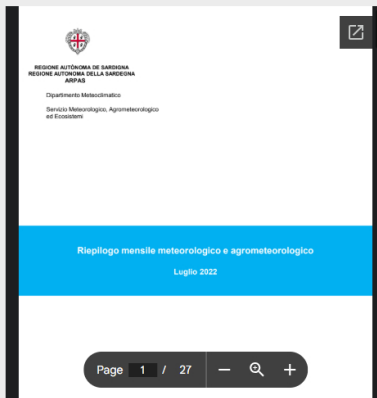
Mentee: Sonia Mereu

This site preserves my Geo+Women's Legacy previously hosted on StoryMaps. It reflects an exploratory phase between geography, visualization, and code, where I tested what could be done with Google's ecosystem.

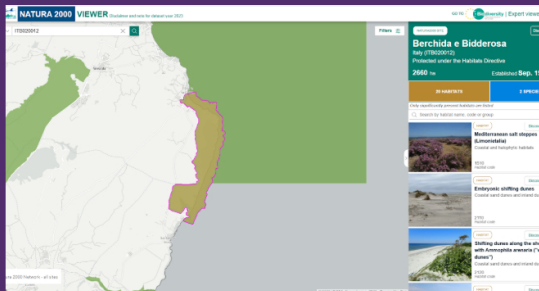
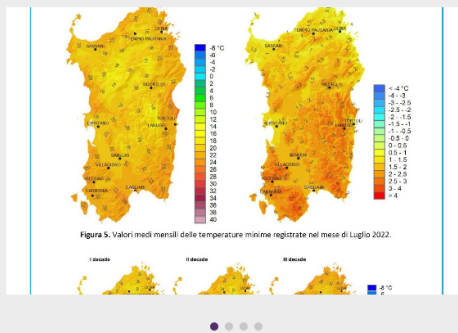
Introduction

The Area of Interest, Capo Comino, is located in Sardinia, Italy, and encloses peculiar environmental specifics (Dunal System, Wetland areas, and Mediterranean maquis).

The summer of 2022 experienced a heatwave, the extent of which we aim to explore using Remote Sensing and compare with data collected from the closest meteorological station (Siniscola) during July 2022 (Riepilogo mensile meteorologico e agrometeorologico Luglio 2022).



Monthly Meteorological and Agrometeorological Summary – July 2022



Natura 2000: The European Network of Natural Areas - Interactive Viewer

The Challenge in Capo Comino

These environments are crucial for the biodiversity they support, including endemic species and migratory birds that use the wetlands as stopover sites.

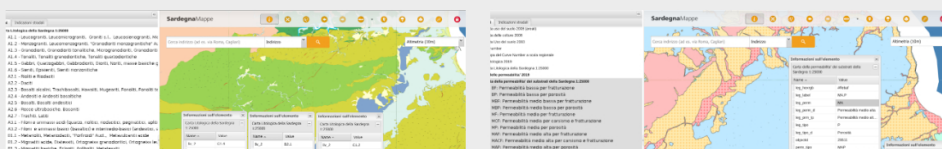
The Mediterranean maquis, a dense scrubland found in the region, further contributes to the area's ecological diversity, providing habitat for various plant and animal species.

The area is included in the European Network of Natural Areas, Natura 2000, which provides a user-friendly map viewer for exploring habitats and species.

Given the ecological significance of this area, understanding the impact of temperature variations, especially during extreme weather events such as heatwaves, is vital (https://biodiversity.europa.eu/habitats/ANNEX1_2110).

Remote sensing techniques, utilizing satellite imagery and aerial photography, offer a comprehensive tool for efficiently monitoring these changes.

Methodological Approach





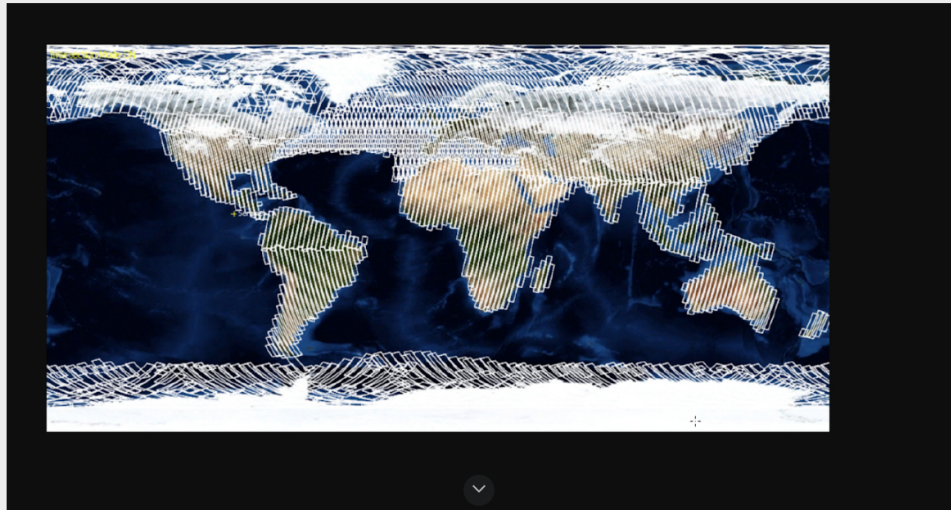
Lithological Map

The Lithological Map shows the distribution and types of rocks at the Earth's surface, helping to understand an area's geological structure.

Permeability Map

The Substrate Permeability Map indicates how easily fluids can pass through the substrate materials, which is crucial for applications such as groundwater studies, contamination spread, and agricultural planning.

Copernicus - European Space Agency - Sentinel Satellites



What's Remote Sensing?

Remote Sensing is the process of measuring or acquiring Information or Properties of an object or phenomenon by a recording device that is not in physical or intimate contact with the object under study.

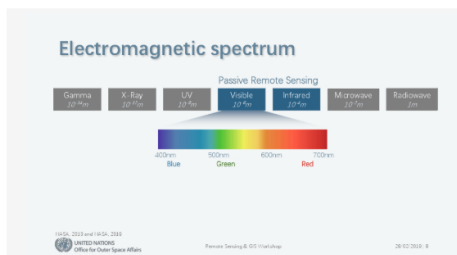
Remote Sensing technologies rely on detecting and analysing electromagnetic energy emitted or reflected by objects on Earth's surface or in its atmosphere.

The electromagnetic spectrum encompasses a wide range of wavelengths and frequencies, from the shorter wavelengths of gamma rays and X-rays to the longer wavelengths of microwaves and radio waves.

Each segment of this spectrum has unique characteristics and interacts differently with the Earth's surface and atmosphere. By analyzing how various materials absorb, reflect, or emit these different wavelengths, remote sensing technologies can gather critical information about the Earth and its environment.

We will focus on specific parts of the electromagnetic spectrum as we delve deeper into our analysis of the summer 2022 heatwave using Sentinel-1 and Sentinel-2 data.

These satellites capture data in various spectral bands, allowing us to observe and quantify the impact of extreme heat events on the Earth's surface and atmosphere. Understanding the electromagnetic spectrum is essential for interpreting these data accurately and deriving meaningful insights about the heatwave's effects on our Area of Interest.



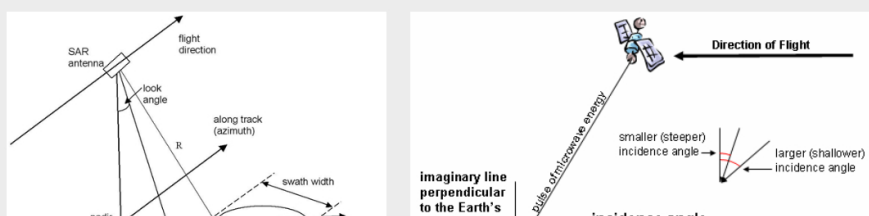
Any Remote Sensing process needs an Energy Source for Reflection: the sensors collect data by detecting the energy reflected or emitted from the Earth's surface.

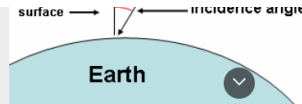
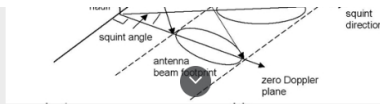
For most sensors operating in the visible and infrared spectrum, sunlight is the primary source of illumination.

In this case we refer to **Passive Remote Sensing**: the sun's energy illuminates the Earth's surface, and the sensors aboard satellites or aircraft measure the amount of light that is reflected back from the surface

Active Remote Sensing and SAR (Synthetic Aperture Radar)

Concepts involved in the Active Remote Sensing





Polarization

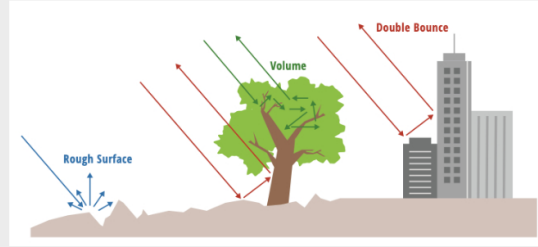
Polarization refers to the orientation of the electromagnetic wave's oscillation.

In the context of SAR remote sensing, polarization is crucial as it significantly impacts the type and quality of data captured. SAR systems can transmit and receive signals in different polarizations, such as horizontal (H) or vertical (V).

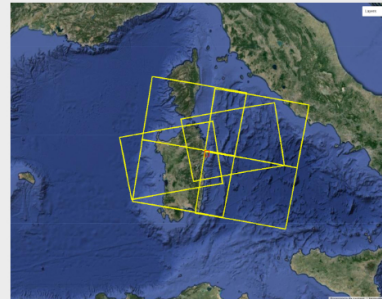
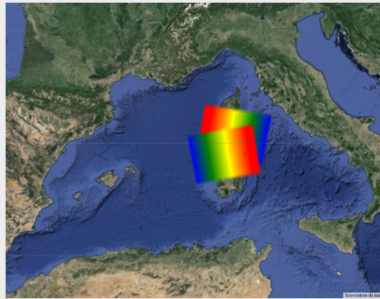
The typical polarizations recorded are HH, VV, HV, and VH, where the first letter represents the transmit polarization and the second the receive polarization.

Each polarization configuration interacts differently with the Earth's surface, providing unique information.

For instance, VV polarization is often more sensitive to water and smooth surfaces, while HV or VH polarizations are better suited for capturing vegetation structure and urban environments.



Scattering mechanisms. Rough surfaces give bright returns due to the wide scattering. Vegetated surfaces cause volumetric scattering, which provides a darker return to the imaging platform. Double bounce returns, found mostly in urban areas, provide the brightest return, as the majority of the energy is redirected back towards the sensor. Credits: [the Sar Handbook](#).



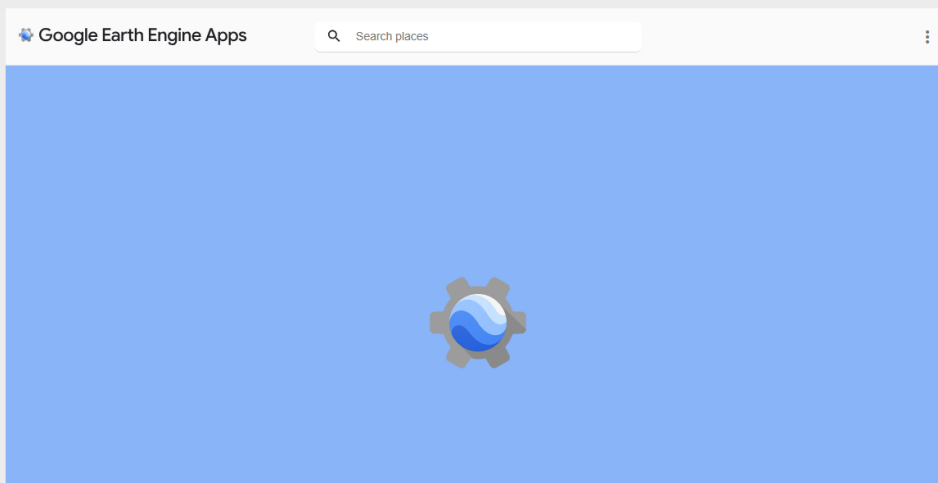
Google Earth Engine Applications

The Google Earth Engine applications presented above are functional prototypes built between 2022 and 2023. They explore different dimensions of the same event, the 2022 heatwave, through radar backscatter, land cover classification, and spectral indices.

While their interfaces remain simple, they illustrate a personal challenge: making satellite data readable for non-experts.

Sentinel 1 Footprint and Incident Angle

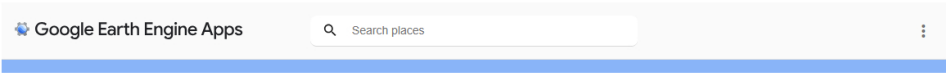
About the App.



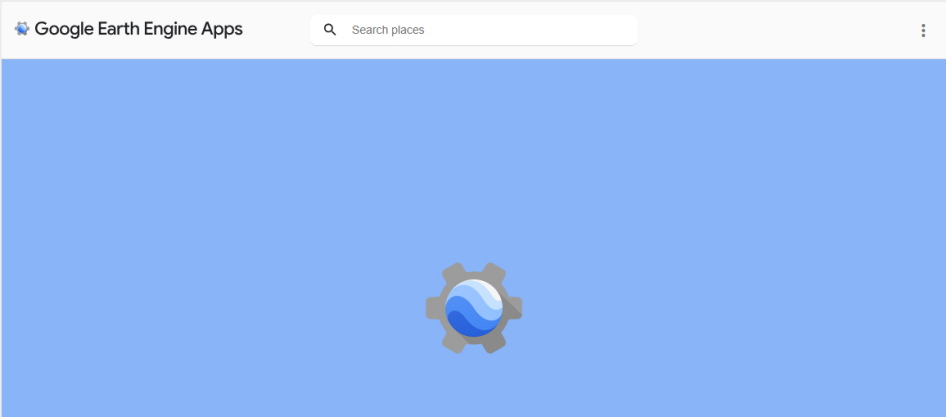
Sentinel 1 Backscattered Profile



Sentinel 2 and Dynamic World



Sentinel 2 Temporal and Bands Exploration

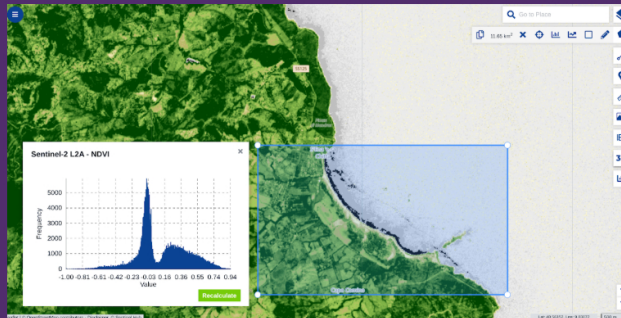


Sentinel 1 LandClass and Remapping

[About the App.](#)

Explore the outputs gallery for charts and plots generated with GEE.

The charts, boxplots, and time profiles shown here demonstrate how GEE can directly render statistical insights, bridging data science and environmental storytelling.



Be Curious

This story is a testament to my journey from a curious SQL and BigQuery beginner to a skilled practitioner proficient in Google Earth Engine (GEE) and the Copernicus platform for geospatial data analysis and applications. It's a journey that involves overcoming challenges, finding solutions with a basic hardware setup, and developing simple apps for a non-expert audience. Although not every experiment reached its conclusion, the process itself shows how curiosity can bridge disciplines. Revisiting it one year later, I see not a collection of GEE Apps, but a record of learning and a sandbox of possibility.